

KELVIN CECIL NEWMAN

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EDUCATION

Bachelor of Arts in Statistics and Economics (Double Major)

Jan. 2021 – Sept. 2024

University of Ghana, Legon

Accra, Ghana

EXPERIENCE

Data Quality Analyst

Oct. 2024 – Sept. 2025

Ghana Statistical Service

Accra, Ghana

- Audited unemployment rate estimates spanning 22 years (1999–2021) across three national survey datasets, reconciling conflicting figures and aligning outputs with IMF and World Bank standards.
- Built a predictive economic model in R with 92% accuracy on Ghana's 2023 Income & Expenditure Survey, directly supporting national income projection reports.
- Developed a text classification model to automate occupation and industry coding — adopted by the Ghana Statistical Service for all future survey cycles, removing manual interpretation errors.
- Validated the full GLSS8 questionnaire post-collection — auditing skip logic, constraint enforcement, and interviewer-coded responses to verify accuracy throughout.
- Produced a structured data inconsistency report outlining cleaning recommendations, submitted to the senior analysis team ahead of the national reporting phase.

Premium Administration Intern — Financial Operations

July 2023 – Sept. 2023

Enterprise Life Assurance

Accra, Ghana

- Audited client premium accounts in PAP, identifying and resolving financial discrepancies — overcharges, missed deductions, and duplicate policies — contributing to a 15% reduction in billing errors.
- Investigated a premium misallocation case where funds from a secondary policy were incorrectly applied to an EduCare policy — traced the source, escalated through compliance channels, and confirmed accurate reconciliation.
- Flagged abandoned and lapsed policies and coordinated follow-ups with underwriters and sales agents, improving portfolio data accuracy and reducing processing errors.
- Conducted client outreach to explain insurance products and facilitate enrolments, contributing to increased policy uptake over the internship period.

PROJECTS

Sika — FX & Commodity Trend Prediction System | *Python, Scikit-learn, TA-lib*

github.com/sika

- Built an end-to-end ML pipeline to forecast daily market direction across forex, crypto and commodity pairs (BTC/USD, GBP/USD, XAU/USD) — achieving 80–90% accuracy using a Multi-Layer Perceptron with L-BFGS optimisation.
- Engineered 7 technical indicators (MACD, RSI, ADX, Stochastic RSI, TTM Trend) as model features, capturing momentum, trend strength, and reversal signals relevant to live trading decisions.
- Designed a modular CLI system with automated model persistence, retraining workflows, and accuracy logging — enabling continuous performance monitoring as market conditions evolve.

SKILLS

Finance & Analysis: FX Market Analysis, Economic Forecasting, Macroeconomic Research, Investment Analysis, Budgeting & Expense Analysis

Programming: Python, R, SQL

Data & Analytics Tools: Scikit-learn, Pandas, TA-Lib, Excel

Reporting & Productivity: Microsoft Office Suite, Git

Domain Knowledge: Macroeconomic & Survey Research, Insurance & Premium Administration, FX & Commodity Markets, IMF/World Bank Reporting Standards

References available upon request.